Shrikaran P

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Professional Summary

Computer Science Engineering student with a strong foundation in Python, Java, Data Structures & Algorithms, Object-Oriented Programming, SQL, and software development. Hands-on experience in AI/ML, REST API development, ETL pipelines, cloud fundamentals, and research-driven projects involving Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG). Passionate about building scalable, data-driven solutions with strong analytical, problem-solving, and collaborative skills.

Technical Skills

- **Programming Languages**: Python, Java, C, C++
- **Core Concepts**: Data Structures & Algorithms, Object-Oriented Programming (OOP)
- **Web Development**: HTML, CSS
- **Databases & Version Control:** SQL, MySQL, Git, GitHub
- **Backend:** FastAPI, REST APIs
- **Cloud & Tools:** AWS, Firebase, Google Colab
- **AI/ML**: Machine Learning, Deep Learning, PyTorch, Scikit-learn,CNN-LSTM
- **Soft Skills**:Team Collaboration ,Quick Learner, Time Management , Leadership, Communication

Educational Qualifications

B.Tech in Computer Science Engineering(6th semester) May 2023 - May 2027
SRM Institute of Science and Technology - Tiruchchirappalli
CGPA - 9.3/10

Achievements

- **3rd Prize Winner (Cash Award)** – Ideathon, SRM Institute of Science and Technology, for developing a Centralized Dam Management System.
- **Participant** – 24-Hour Hackathon, Dhanalakshmi Srinivasan University, where I developed a Farmer Market Price Prediction System as part of a team.
- **Foundations of AI/ML – GeeksforGeeks -** Gained hands-on experience in data preprocessing, model building, and evaluation techniques
- **Java Full Stack Development – EduSkil**l - Built web-based applications with structured data handling and backend integration
- **AWS Academy Cloud Foundations – AWS**
  Learned fundamentals of cloud computing and data storage for scalable data processing

Research Internship(NIT Trichy)

- Motor Imagery EEG Classification (BCI) — Worked with BCI Competition IV Dataset 2a and BCI Competition IV Dataset 2b EEG .gdf recordings for motor imagery analysis.
- Preprocessed raw EEG signals, selected motor cortex channels, and generated STFT-based time–frequency spectrogram representations.
- Developed CNN-based models for left/right hand motor imagery classification and subject-wise performance evaluation.

Projects

A Secure On-Premises ETL Pipeline Using OCR and Local LLMs

- Designed a secure ETL pipeline to extract text from scanned documents using OCR and transform unstructured data into structured formats with Local LLMs.
- Automated document processing, data validation, and information extraction while ensuring data privacy and improving processing efficiency.

Contrastive Knowledge Graph-RAG for Concept Misconceptions

- Engineered a Knowledge Graph-based Retrieval-Augmented Generation (KG-RAG) system using contrastive learning to detect and explain concept misconceptions.
- Integrated embeddings, vector retrieval, and Large Language Models to improve retrieval accuracy, contextual understanding, and explainable AI responses.

Weather Data Pipeline API

- Built a FastAPI-based REST API to ingest real-time weather data from the OpenWeather API, process it, and store it in PostgreSQL for efficient data management.
- **Implemented** secure API endpoints with filtering, validation, and historical weather retrieval, enabling reliable data access for analytics and reporting.

Intrusion Detection System using CICIDS2017

- Designed a deep learning–based intrusion detection model to classify malicious and benign network traffic using the CICIDS2017 dataset.
- Compared model performance against traditional rule-based methods using accuracy, precision, recall, and F1-score metrics.

Farmer Market Price Prediction System

- Created a machine learning model to predict crop market prices using historical mandi data, seasonal trends, weather conditions, and demand–supply analysis.

- Provided predictive analytics and market insights to support informed selling decisions and improve farmers' profitability.

Smart Interconnected Water Dam Alert System

- Designed and developed an AI-enabled interconnected dam monitoring system using IoT sensors for real-time water level monitoring, flood prediction, and automated early warning alerts.
- Integrated cloud-based data analytics, predictive modeling, and interactive dashboards to improve disaster preparedness and support data-driven water resource management.

Research & Publications

- A Secure On-Premises ETL Pipeline Using OCR and Local LLMs - ICCMC 2025, Tiruppur
- Contrastive Knowledge Graph-RAG for Concept Misconceptions-RECCAP 2026, IIT Palakkad